WIA1005 Network Technology Foundation

Chapter 8

Switching Concept

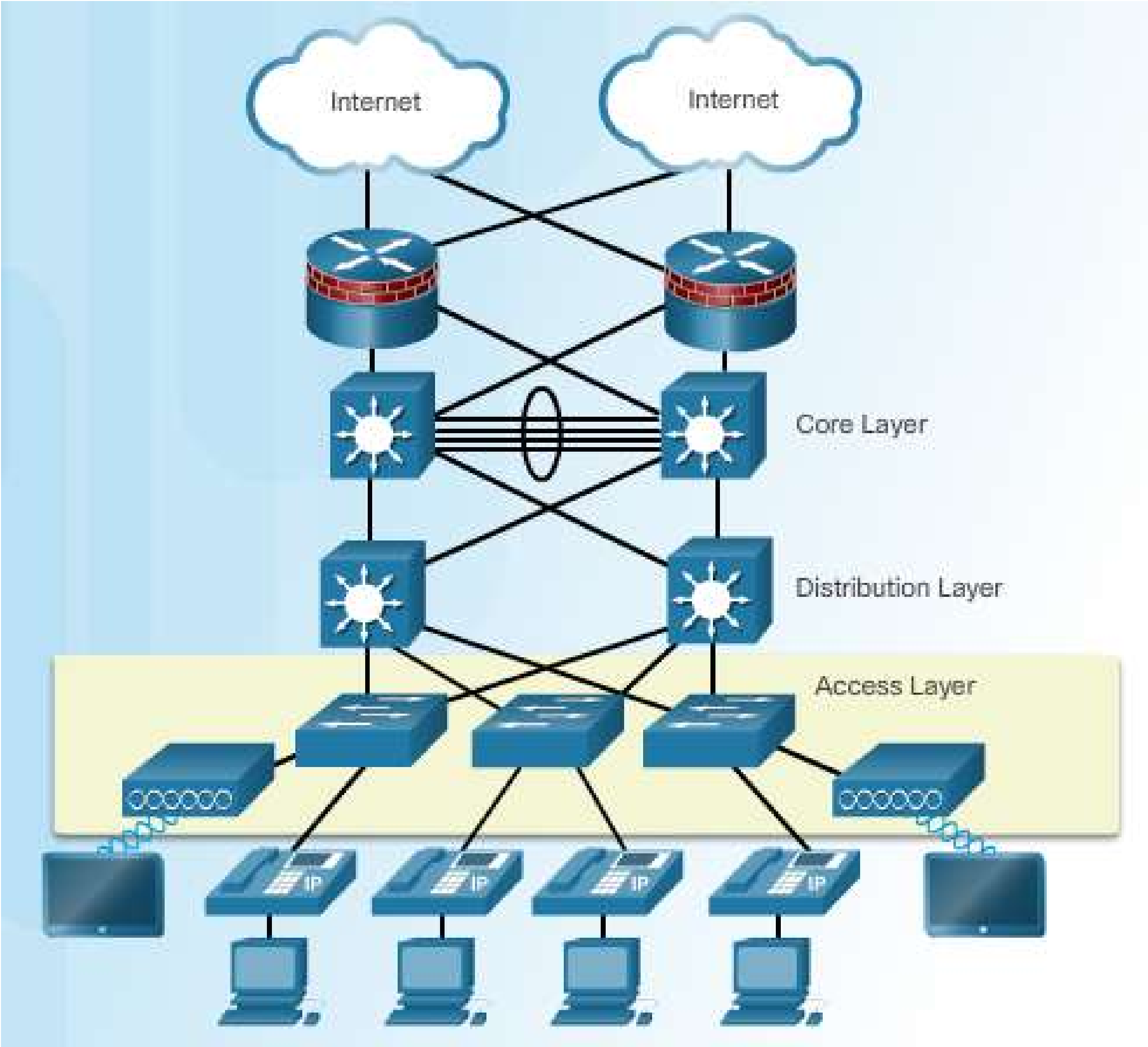
Contents

- Introduction
- Switching Domain
- Switch Configuration
- Switch Security Management

Introduction

- **Access Layer** - represents the network edge, where the primary function of an access layer switch is to provide network access to the user.
- **Distribution Layer** – the interfaces between the access layer and the core layer. Providing high availability through redundant distribution layer switches.
- **Core Layer** – The network backbone. It connects several layers of the campus network. The core layer serves as the aggregator for all of the other campus blocks and ties the campus together with the rest of the network.

Introduction



Introduction

- The decision on how a switch forwards traffic is made based on the flow of that traffic.
  - **Ingress** - This is used to describe the port where a frame enters the device.
  - **Egress** - This is used to describe the port that frames will use when leaving the device.
- A LAN switch forwards traffic based on the ingress port and the destination MAC address of an Ethernet frame.
- Switch has a MAC address table that stored in content addressable memory (CAM) which is a special type of memory used in high-speed searching applications.

Introduction

- Step 1. Learn - Examining the Source MAC Address
- Every frame that enters a switch is checked for new information to learn. It does this by examining the **source MAC address** of the frame and port number where the frame entered the switch:
  - If the source MAC address does not exist in the MAC address table, the MAC address and incoming port number are added to the table.
  - If the source MAC address does exist, the switch updates the refresh timer for that entry. By default, most Ethernet switches keep an entry in the table for five minutes.

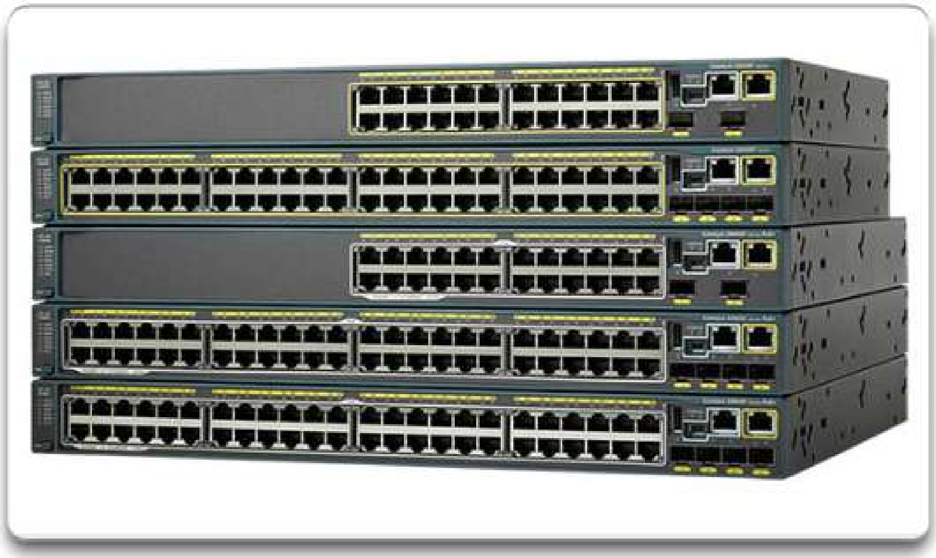
Introduction

- Step 2. Forward - Examining the Destination MAC Address
- If the destination MAC address is a unicast address, the switch will look for a match between the destination MAC address of the frame and an entry in its MAC address table:
  - If the destination MAC address is in the table, it will forward the frame out of the specified port.
  - If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an unknown unicast.
- If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

Introduction

- **Fixed Configuration Switches**
  - Fixed configuration switches do not support features or options beyond those that originally came with the switch.

Fixed Configuration Switches



Introduction

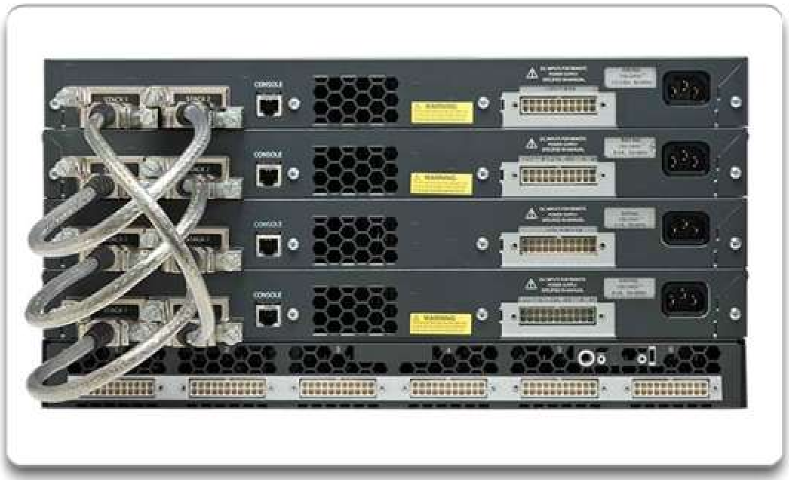
- **Modular Configuration Switches**
  - Modular configuration switches typically come with different sized chassis that allow for the installation of different numbers of modular line cards .

Modular Configuration Switches

Introduction

- **Stackable Configuration Switches**
  - Stackable configuration switches can be interconnected using a special cable that provides high-bandwidth throughput between the switches.
  - The stacked switches effectively operate as a single larger switch. Stackable switches are desirable where fault tolerance and bandwidth availability are critical.

Stackable Configuration Switches



Introduction

- A switch stack can consist of up to nine switches connected through their **StackWise** ports.
- One switch called the **stack master** controls the operation of the stack. Switch stacks are managed using a **single IP address**. The master contains the saved and running configuration files for the stack and each member has a current copy of these files for backup purposes.
- If the master becomes unavailable, there is an automatic process to elect a new master from the remaining stack members. The highest stack-member **priority value** will become the master.

Introduction

- Every switch is uniquely identified by its own stack member number. The first number after the interface-type is the stack-member number.

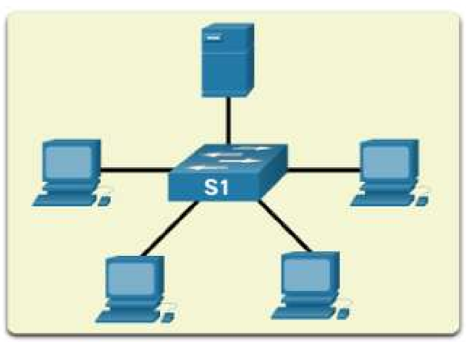
```
Switch# show running-config | begin interface
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
<output omitted>
!
interface GigabitEthernet1/0/52
!
interface GigabitEthernet2/0/1
!
interface GigabitEthernet2/0/2
!
<output omitted>
!
interface GigabitEthernet2/0/52
!
interface GigabitEthernet3/0/1
!
interface GigabitEthernet3/0/2
!
<output omitted>
!
interface GigabitEthernet3/0/52
!
interface GigabitEthernet4/0/1
!
interface GigabitEthernet4/0/2
!
<output omitted>
!
interface GigabitEthernet4/0/52
```

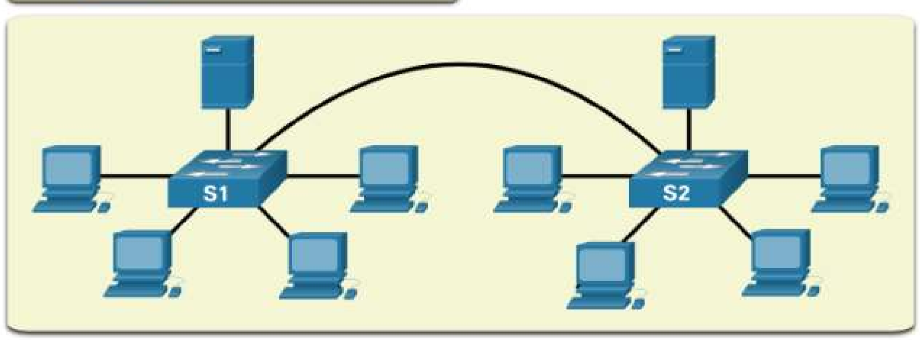
Switching Domain

- Ethernet switch ports will auto-negotiate full-duplex when the adjacent device can also operate in full-duplex.
- If the switch port is connected to a device operating in half-duplex, such as a legacy hub, then the switch port will operate in half-duplex.
- If an Ethernet switch port is operating in half-duplex, each segment is in its own collision domain. There are no collision domains when switch ports are operating in full-duplex.

Switching Domain

- A collection of interconnected switches forms a single broadcast domain. This broadcast domain is referred to as the MAC broadcast domain. The MAC broadcast domain consists of all devices on the LAN that receive broadcast frames from a host.





Switching Domain

- Characteristics of switches :
  - **Fast port speeds**
    - Most access layer switches support 100 Mbps and 1 Gbps port speeds.
    - Distribution layer switches support 100 Mbps, 1 Gbps, and 10 Gbps port speeds
    - Core layer and data center switches may support 100 Gbps, 40 Gbps, and 10 Gbps port speeds.
  - **Fast internal switching**
    - Switches use a fast internal bus or shared memory to provide high performance.

Switching Domain

- **Large frame buffers**
  - Switches use large memory buffers to temporarily store more received frames before having to start dropping them. This enables ingress traffic from a faster port (e.g., 1 Gbps) to be forwarded to a slower (e.g., 100 Mbps) egress port without losing frames.
- **High port density**
  - A high port density switch lowers overall costs because it reduces the number of switches required. For instance, if 96 access ports were required, it would be less expensive to buy two 48-port switches instead of four 24-port switches.

Switch Configuration

- First, the switch loads a **power-on self-test** (POST) program stored in ROM. POST checks the CPU subsystem. It tests the CPU, DRAM, and the portion of the flash device that makes up the flash file system.
- Next, the switch loads the **boot loader** software. The boot loader is a small program stored in ROM that is run immediately after POST successfully completes. The boot loader performs low-level CPU initialization. It initializes the CPU registers, which control where physical memory is mapped, the quantity of memory, and its speed.

Switch Configuration

- The boot loader initializes the flash file system on the system board.
- Finally, the boot loader locates and loads a default IOS operating system software image into memory and gives control of the switch over to the IOS. The IOS operating system then initializes the interfaces using the commands found in the configuration file, startup configuration, which is stored in NVRAM.

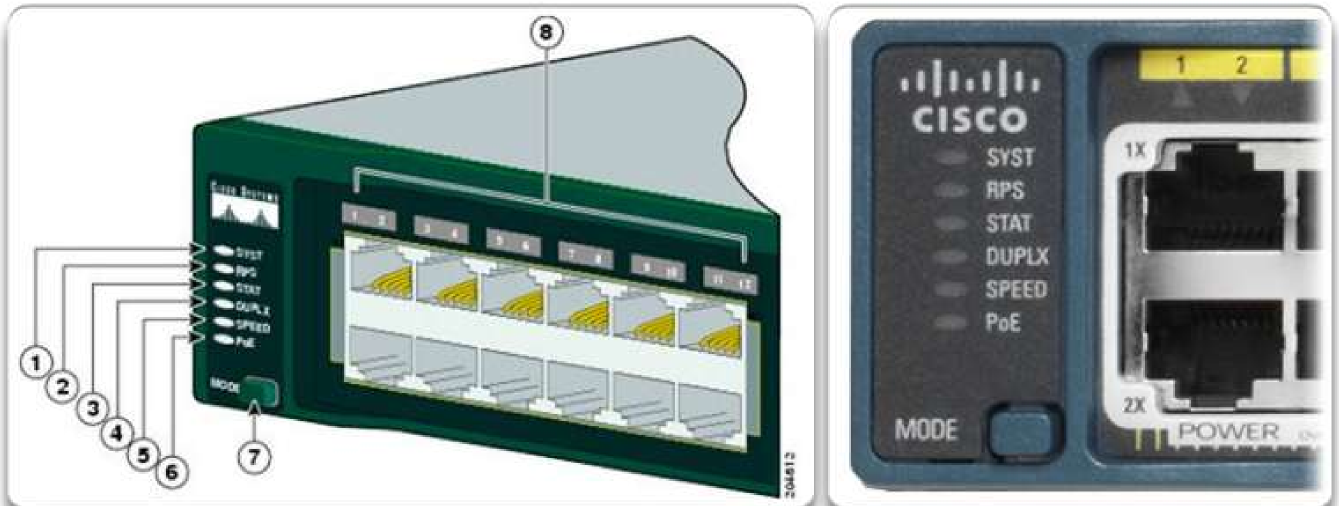
```
S1(config)# boot system flash:/c2960-lanbasek9-mz.150-2.SE/c2960-lanbasek9-
mz.150-2.SE.bin
```

Switch Configuration

- The switch have several status LED indicator lights.
- **System LED**
  - LED off - System is OFF.
  - LED is green - System is operating normally.
  - LED is amber - System is not functioning properly.
- Redundant Power System (RPS) LED
- Port Status LED
- Port Duplex LED
- Port Speed LED
- Power over Ethernet (PoE) Mode LED

Switch Configuration

Switch LEDs



Catalyst 2960 Switch LEDs

1	The system LED	5	The port speed LED
2	The RPS LED (if RPS is supported on the switch)	6	The PoE status LED (if PoE is supported on the switch)
3	The port status LED (This is the default mode.)	7	The Mode button
4	The port duplex mode LED	8	The port LEDs

Switch Configuration

- How to access boot loader?
  - Step 1. Connect a PC by console cable to the switch console port. Configure terminal emulation software to connect to the switch.
  - Step 2. Unplug the switch power cord.
  - Step 3. Reconnect the power cord to the switch and, within 15 seconds, press and hold down the Mode button while the System LED is still flashing green.
  - Step 4. Continue pressing the Mode button until the System LED turns briefly amber and then solid green; then release the Mode button.
  - Step 5. The boot loader **switch:** prompt appears in the terminal emulation software on the PC.

Switch Configuration

```
switch: set
BOOT=flash:/c2960-lanbasek9-mz.122-55.SE7/c2960-lanbasek9-mz.122-55.SE7.bin
(output omitted)
switch: flash_init
Initializing Flash...
flashfs[0]: 2 files, 1 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 32514048
flashfs[0]: Bytes used: 11838464
flashfs[0]: Bytes available: 20675584
flashfs[0]: flashfs fsck took 10 seconds.
...done Initializing Flash.

switch: dir flash:
Directory of flash:/
    2  -rwx  11834846                 c2960-lanbasek9-mz.150-2.SE8.bin
    3  -rwx  2072                     multiple-fs

switch: BOOT=flash:c2960-lanbasek9-mz.150-2.SE8.bin
switch: set
BOOT=flash:c2960-lanbasek9-mz.150-2.SE8.bin
(output omitted)
switch: boot
```

Switch Configuration

- To prepare a switch for remote management access, the switch must be configured with an **IP address, subnet mask and the default gateway**.
- The **switch virtual interface (SVI)** should be assigned an IP address.
- By default, the switch is configured to have the management of the switch controlled through VLAN 1.

```
S1#configure terminal
S1(config)#interface vlan 1
S1(config-if)#ip address 172.17.99.11 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#ip default-gateway 172.17.99.1
```

Switch Configuration

- For security purposes, it is considered a best practice to use a VLAN other than VLAN 1 for the management VLAN.
- In order to configure IPv6 on the Switch (2960), you will need to enter **sdm prefer dual-ipv4-and-ipv6 default** and then reload the switch.

```
S1(config)# interface vlan 99
S1(config-if)# ip address 172.17.99.11 255.255.255.0
S1(config-if)# ipv6 address 2001:db8:acad:99::1/64
S1(config-if)# no shutdown
```

Switch Configuration

- Switch ports can be manually configured with specific **duplex and speed** settings.
- The **duplex** command accepts auto, full and half.
- The **speed** command accepts the speed in Mbps.
- We can enable **auto-MDIX** on the switch ports. When auto-MDIX is enabled, the interface automatically detects the required cable connection.

```
S1(config)# interface FastEthernet0/1
S1(config-if)# duplex full
S1(config-if)# speed 100
S1(config-if)# mdix auto
```

Switch Configuration

Task	IOS Commands
Display interface status and configuration.	`S1# show interfaces [interface-id]`
Display current startup configuration.	`S1# show startup-config`
Display current running configuration.	`S1# show running-config`
Display information about flash file system.	`S1# show flash`
Display system hardware and software status.	`S1# show version`
Display history of command entered.	`S1# show history`

Display IP information about an interface.	`S1# show ip interface [interface-id]`
	OR
	`S1# show ipv6 interface [interface-id]`

Display the MAC address table.	`S1# show mac-address-table`
	OR
	`S1# show mac address-table`

Switch Security Management

- **Secure Shell (SSH)** is a protocol that provides a secure management connection to a remote device.
- Telnet is an older protocol that uses insecure plaintext transmission of both the login authentication (username and password) and the data transmitted between the communicating devices.
- SSH provides security for remote connections by providing strong encryption when a device is authenticated (username and password) and also for the transmitted data between the communicating devices.
- SSH is assigned to TCP port 22. Telnet is assigned to TCP port 23.

Switch Security Management

- **Configuring SSH**
  - Configure the hostname.
  - Configure the IP domain name.
  - Generate RSA key pairs - A minimum modulus size of 1,024 bits is required.
  - Configure user authentication.
  - Configure the vty lines.
- Enabling SSH version 2
  - ip ssh version 2
  - show ip ssh

Switch Security Management

Configure SSH for Remote Management


```
S1# configure terminal
S1(config)# ip domain-name cisco.com
S1(config)# crypto key generate rsa
The name for the keys will be: S1.cisco.com
...
How many bits in the modulus [512]: 1024
...
S1(config)# username admin password ccna
S1(config)# line vty 0 15
S1(config-line)# transport input ssh
S1(config-line)# login local
S1(config)# end
```

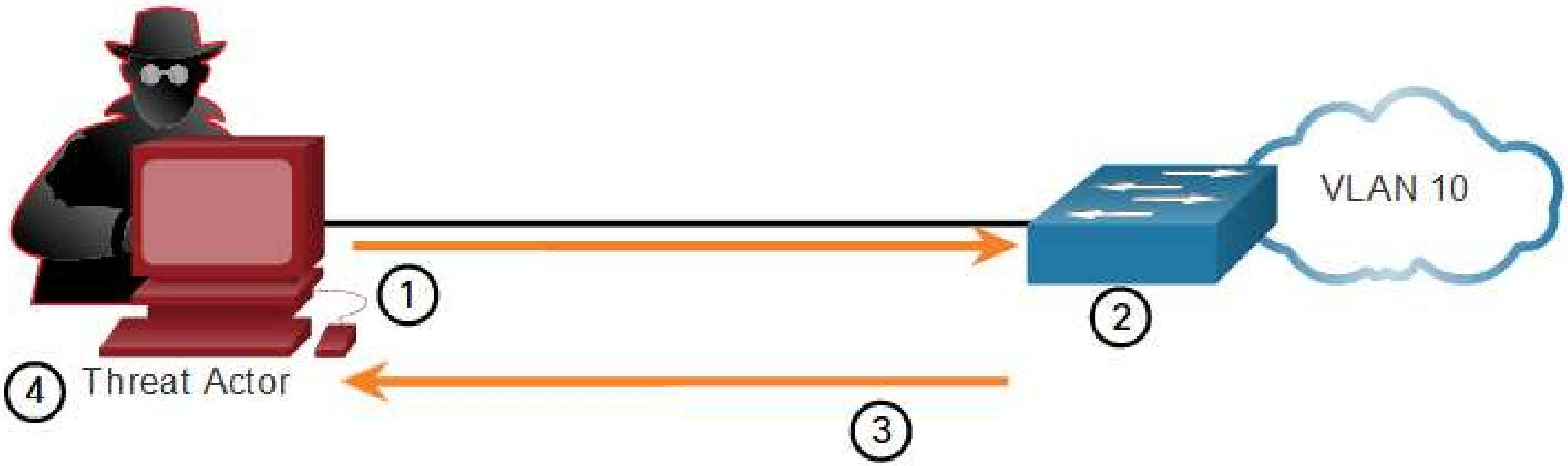
Switch Security Management

- **MAC Address Flooding**
  - The MAC address table in a switch contains the MAC addresses associated with each physical port and the associated VLAN for each port.
  - When a Layer 2 switch receives a frame, the switch looks in the MAC address table for the destination MAC address.
  - If the MAC address does not exist in the MAC address table, the switch floods the frame out of every port on the switch, except the port where the frame was received.

Switch Security Management

- MAC address tables are limited in size. MAC flooding attacks make use of this limitation to overwhelm the switch with fake source MAC addresses until the switch MAC address table is full. (the **macof** attacking tool)
- When this occurs, the switch treats the frame as an unknown unicast and begins to flood all incoming traffic out all ports on the same VLAN without referencing the MAC table.
- This condition allows a threat actor to capture all of the frames sent from one host to another on the local LAN or local VLAN.

Switch Security Management

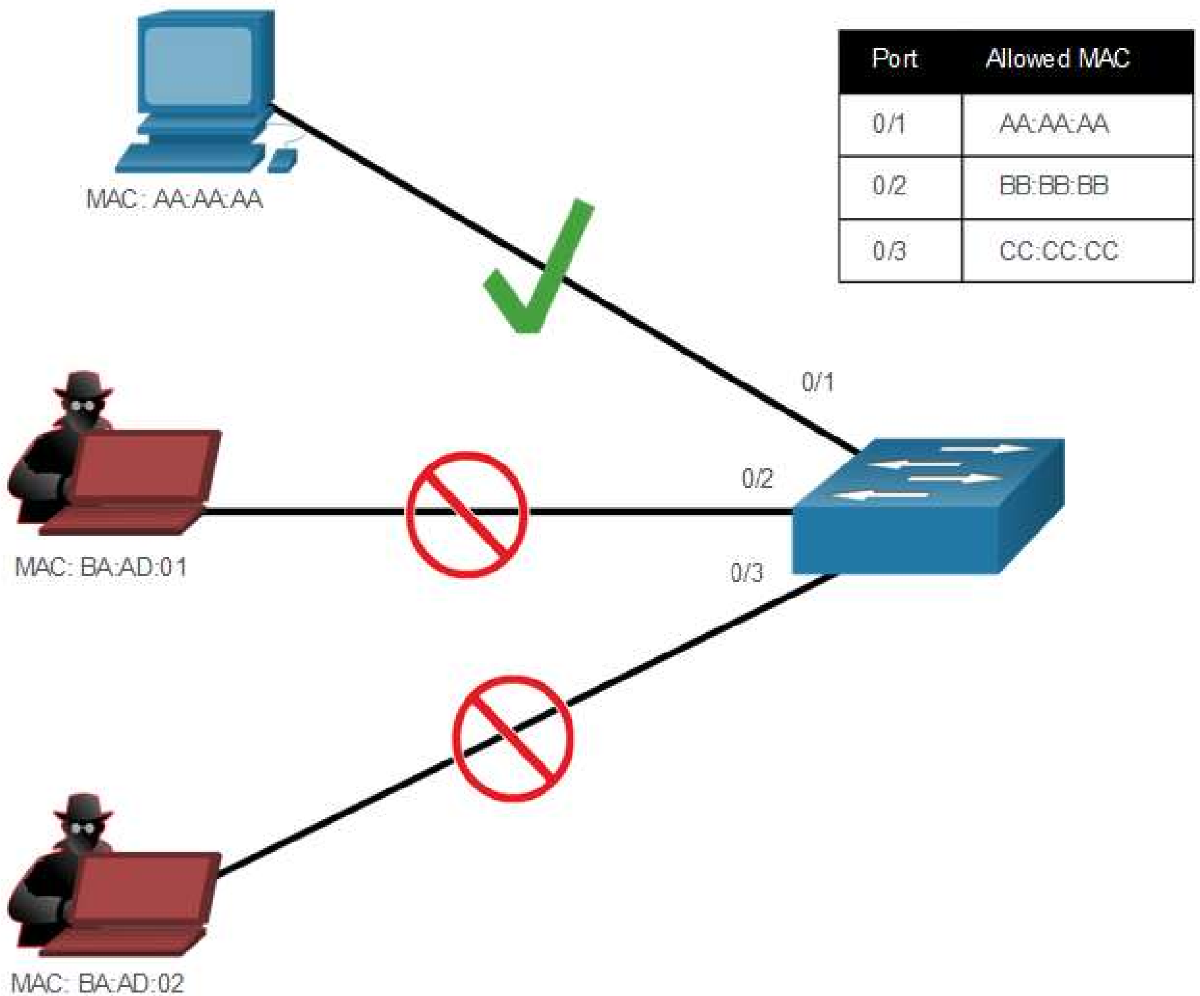


1. The threat actor is connected to VLAN 10 and uses **macof** to rapidly generate many random source and destination MAC and IP addresses.
2. Over a short period of time, the switch's MAC table fills up.
3. When the MAC table is full, the switch begins to flood all frames that it receives. As long as **macof** continues to run, the MAC table remains full and the switch continues to flood all incoming frames out every port associated with VLAN 10.
4. The threat actor then uses packet sniffing software to capture frames from any and all devices connected to VLAN 10.

Switch Security Management

- All switch ports should be secured using port security. **Port security** limits the number of valid MAC addresses allowed on a port.
- By limiting the number of permitted MAC addresses on a port to one, port security can be used to control unauthorized access to the network
- If a port is configured as a secure port and the maximum number of MAC addresses is reached, any additional attempts to connect by unknown MAC addresses will generate a security violation.
- A security violation also occurs when an address learned or configured on one secure interface is seen on another secure interface in the same VLAN.

Switch Security Management



Port	Allowed MAC
0/1	AA:AA:AA
0/2	BB:BB:BB
0/3	CC:CC:CC

Switch Security Management

- There are a number of ways to configure port security.
- **Static secure MAC addresses** - MAC addresses that are manually configured on a port.
- **Dynamic secure MAC addresses** - MAC addresses that are dynamically learned and stored only in the address table. MAC addresses configured in this way are removed when the switch restarts.
- **Sticky secure MAC addresses** - MAC addresses that can be dynamically learned or manually configured and then stored in the address table and added to the running configuration.

Switch Security Management

- An interface can be configured for one of three violation modes.
- **Protect** - When violation occurs, packets with unknown source addresses are dropped until a sufficient number of secure MAC addresses are removed, or the number of maximum allowable addresses is increased. There is **no notification** that a security violation has occurred.
- **Restrict** - When violation occurs, packets with unknown source addresses are dropped until a sufficient number of secure MAC addresses are removed, or the number of maximum allowable addresses is increased. In this mode, there is **a notification** that a security violation has occurred.

Switch Security Management

- **Shutdown** - In this (default) violation mode, a port security violation causes the interface to immediately become **error-disabled and turns off the port** LED. It increments the violation counter.

Security Violation Modes

Violation Mode	Forwards Traffic	Sends Syslog Message	Displays Error Message	Increases Violation Counter	Shuts Down Port
Protect	No	No	No	No	No
Restrict	No	Yes	No	Yes	No
Shutdown	No	No	No	Yes	Yes

Switch Security Management

```
S1#configure terminal
S1(config)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport port-security
S1(config-if)#switchport port-security mac-address sticky
S1(config-if)#switchport port-security maximum 1
S1(config-if)#switchport port-security violation {protect | restrict |shutdown}
```

To re-enable the port, first use the **shutdown** command, then, use the **no shutdown** command to make the port operational, as shown in the example.

```
S1(config)# interface fa0/18
S1(config-if)# shutdown
*Sep 20  07:11:18.845: %LINK-5-CHANGED: Interface FastEthernet0/18, changed state to
administratively down
S1(config-if)# no shutdown
*Sep 20  07:11:32.006: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to up
*Sep 20  07:11:33.013: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/18,
changed state to up
S1(config-if)#
```

Switch Security Management

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport port-security
S1(config-if)# switchport port-security maximum 4
S1(config-if)# switchport port-security mac-address aaaa.bbbb.1234
S1(config-if)# switchport port-security mac-address sticky
S1(config-if)# end
S1# show port-security interface fa0/1
Port Security               : Enabled
Port Status                 : Secure-up
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses       : 4
Total MAC Addresses         : 1
Configured MAC Addresses    : 1
Sticky MAC Addresses        : 0
Last Source Address:Vlan    : 0000.0000.0000:0
Security Violation Count    : 0
S1# show port-security address
               Secure Mac Address Table
-----------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports   Remaining Age
                                                                   (mins)
----    -----------       ----                          -----   -------------
   1    aaaa.bbbb.1234    SecureConfigured              Fa0/1        -
-----------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 8192
```

Switch Security Management

- Port security aging can be used to set the aging time for static and dynamic secure addresses on a port. It is used to remove secure MAC addresses on a secure port without manually deleting the existing secure MAC addresses.
- Two types of aging:
  - **Absolute** - The secure addresses on the port are deleted after the specified aging time.
  - **Inactivity** - The secure addresses on the port are deleted only if they are inactive for the specified aging time.
- Aging time limits can also be increased to ensure past secure MAC addresses remain, even while new MAC addresses are added.

Switch Security Management

```
S1(config)# interface fa0/1
S1(config-if)# switchport port-security aging time 10
S1(config-if)# switchport port-security aging type inactivity
S1(config-if)# end
S1# show port-security interface fa0/1
Port Security                : Enabled
Port Status                  : Secure-shutdown
Violation Mode               : Restrict
Aging Time                   : 10 mins
Aging Type                   : Inactivity
SecureStatic Address Aging : Disabled
Maximum MAC Addresses        : 4
Total MAC Addresses          : 1
Configured MAC Addresses     : 1
Sticky MAC Addresses         : 0
Last Source Address:Vlan     : 0050.56be.e4dd:1
Security Violation Count     : 1
```